

대한진단검사정도관리협회
Korean Association of External Quality Assessment Service

검사전단계 오류 사례와 검사실 대응 가이드라인

동국대일산병원 정재우

① 검사전단계 오류란

② 관련 사례들

③ 대응 지침들

정리

① 점짠점짠외혁오류란 오류란

② 관련 사례들

③ 대응 지침들

정리

definition of laboratory errors

- ISO/TS 22367
  - failure of planned action to be completed as intended,
  - or use a wrong plan to achieve an aim,



~~ISO/TS 22367:2008~~

Medical laboratories — Reduction of error through risk management and continual improvement

Withdrawn (Edition 1, 2008)

→ New version available: ISO 22367:2020

진단검사의학에서 오류들의 간과되는 이유들

Table 1 Errors in laboratory medicine: reasons for a neglect

1. Heterogeneous and ambiguous definition of what a laboratory error really is;
2. Difficulties in discovering and identifying all types of errors and need for well-designed study protocols aiming to evaluate all steps in the total testing process (TTP);
3. Complexity of TTP and need for cooperation and integration between different health-care providers;
4. Poor perception by physicians and other stakeholders of the harmfulness of errors in laboratory medicine;
5. Laboratory professionals reluctant to report and disclose data on types of errors and their frequency;
6. Increasing use of complementary/alternative testing options (e.g. point-of-care, near-patient and self-monitoring).

Ann Clin Biochem. 2010;47(Pt 2):101-10.

시대별 진단검사의 오류 빈도

The "eras" in discovering errors in laboratory medicine.

Years	Error detection (types)	Milestones
•1947–'90 s The "ancient era"	Analytical errors	Belk et al. [16]
•'90 s	Errors within clinical laboratory walls	Ross et al. [17] Plebani et al. [18]
The "mid era"		Carraro et al. [19]
•2000 The "modern era"	Errors in laboratory medicine	Plebani et al. [20]
	Towards	
•Today	Laboratory errors in diagnostic medicine	

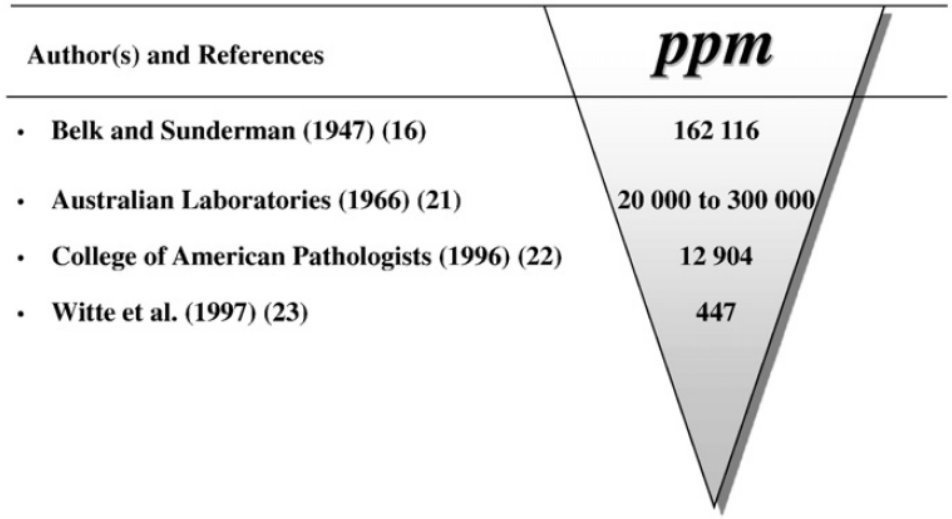
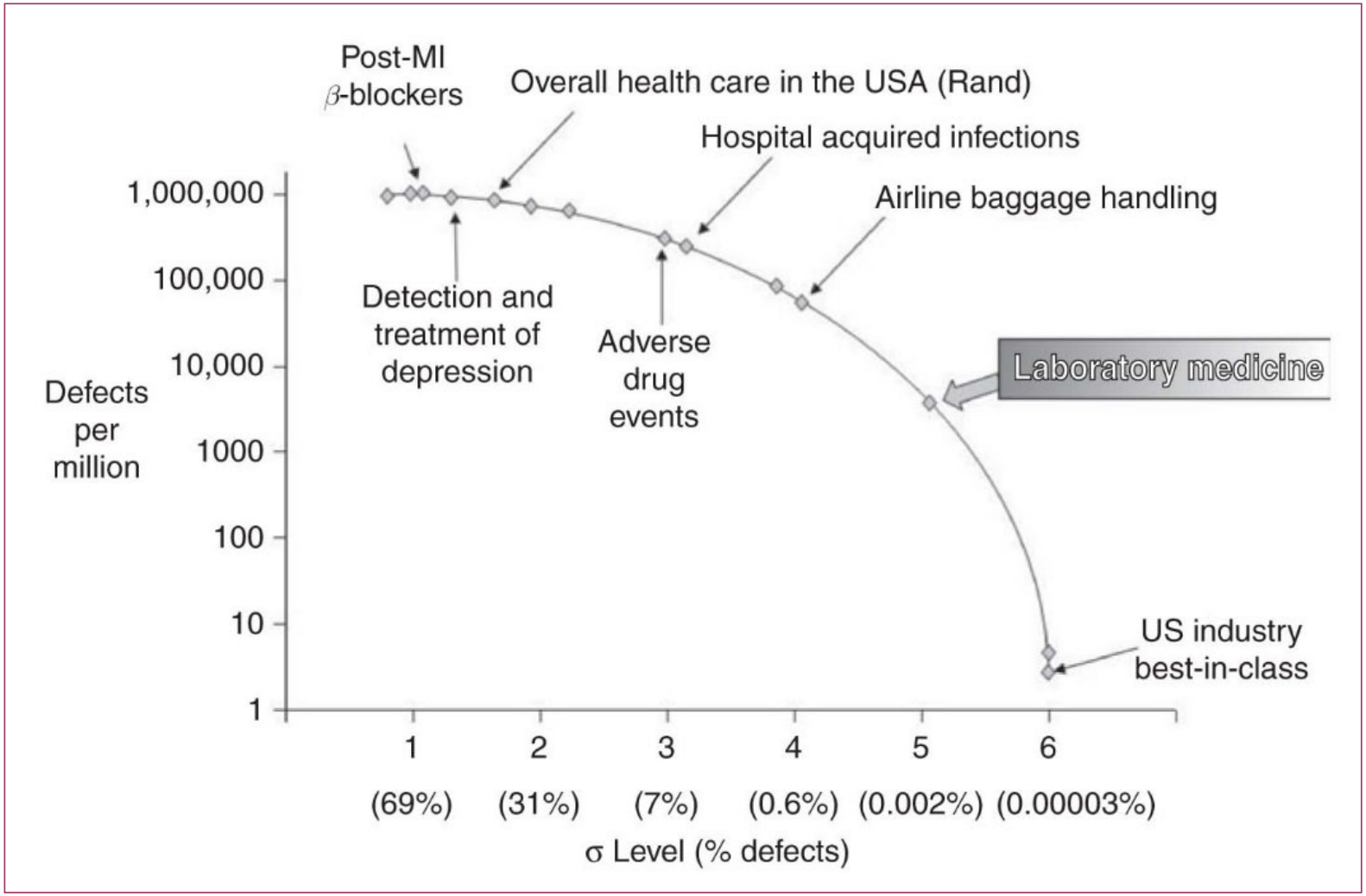


Fig. 1. Decrease in analytical error rates over time.

Clin Chim Acta. 2009;404(1):16-23.

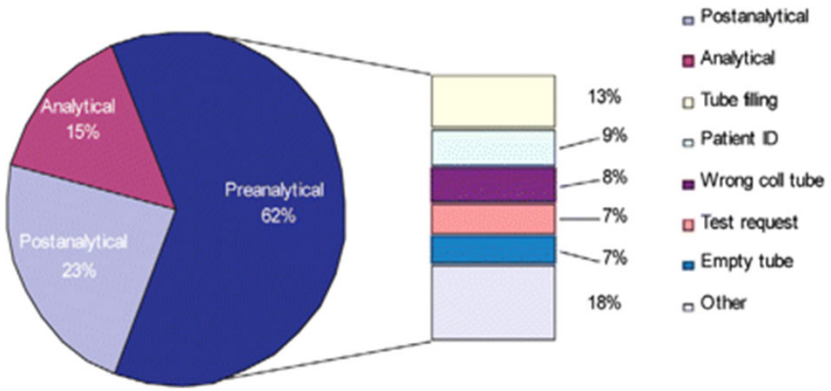


Clin Chim Acta. 2009;404(1):2-5.

응급 검사에서의 오류

Table 1. Laboratory errors in stat testing.

Defects: detection steps	Defects found: No.	Defects found: Frequency, %
Preanalytical		
Specimen collected from infusion route	3	1.9
Sample contaminated	1	0.6
Tube filling error	21	13.1
Empty tube	11	6.9
Inappropriate container	13	8.1
Nonrefrigerated sample	3	1.9
Missing tube	5	3.1
Digoxin test timing error	1	0.6
Patient identification error	14	8.8
Request procedure error	12	7.5
Data communication conflict	6	3.8
Physician's request order missed	3	1.9
Order misinterpreted	2	1.3
Check-in not performed (in the Laboratory Information Systems)	4	2.5
Subtotal	99	61.9
Analytical		
Instrument-caused random error	3	1.9
Analytical inaccuracy not recognized	21	13.1
Subtotal	24	15
Postanalytical		
Results communication breakdown	32	20
Lack of communication within laboratory	3	1.9
TAT excessive	2	1.3
Subtotal	37	23.1



Clin Chem. 2007;53(7):1338-42.

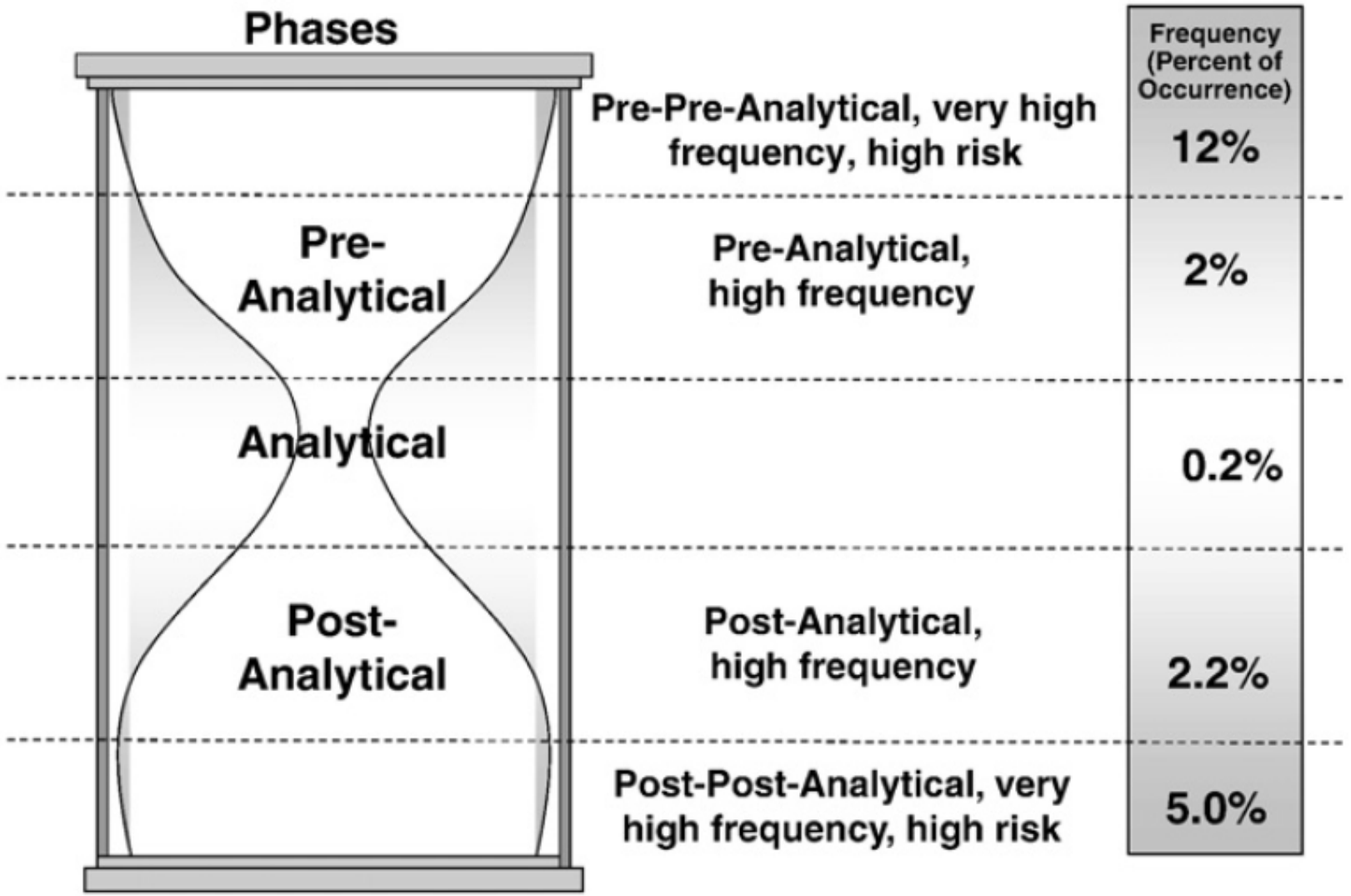
Frequency of errors (%) in the main phases of the total testing process.

Year	Author(s)	Pre	Intra analytic	Post
•1991	Ross et al.	45.5	7.3	47.2
•1997	Plebani et al.	68.2	13.3	18.5
•2003	Astion et al.	71.0	18.0	11.0
•2007	Carraro et al.	61.9	15.0	23.1

Clin Chem. 2007;53(7):1338-42.

	Absolute frequency, ppm	
	1996	**2006**
Total errors	4667	3092
Preanalytical	3186	1913
Analytical	617	464
Postanalytical	864	715

Clin Chim Acta. 2009;404(1):16-23.



Clin Chim Acta. 2009;404(1):16-23.

types and frequency of errors in the phases

Phase of the TTP	Type of error	Relative frequency (%)
Pre-pre-analytical	Inappropriate test request	
	Order entry	
	Patient/specimen misidentification	
	Sample collected from infusion route	
	Sample collection (haemolysis, clotting, insufficient volume, etc.)	46–68.2%
	Inappropriate container	
	Handling, storage and transportation	
Pre-analytical	Sorting and routing	
	Pour-off	
	Aliquoting, pipetting and labelling	3.0–5.3%
	Centrifugation (time and/or speed)	
Analytical	Equipment malfunction	
	Sample mix-ups	7.0–13%
	Interference (endogenous or exogenous)	
	Undetected failure in quality control	
Post-analytic	Erroneous validation of analytical data	
	Failure in reporting/addressing the report	
	Excessive turn-around-time	12.5–20%
	Improper data entry and manual transcription error	
	Failure/delay in reporting critical values	
Post-post-analytic	Delayed/missed reaction to laboratory reporting	
	Incorrect interpretation	
	Inappropriate/inadequate follow-up plan	25–45.5%
	Failure to order appropriate consultation	

Ann Clin Biochem. 2010;47(Pt 2):101-10.

Ann Lab Med 2012;32:5-16

types and frequency of errors in clinical laboratory

Type of error	Frequency	%	Cause of error
Preanalytical			
Hemolyzed sample	508	53.2	Wrong phlebotomy technique, incorrect transport centrifugation before sample is clotted
Insufficient sample	72	7.5	Lack of knowledge regarding sample volume required, difficult sampling as in pediatric and old patients
Lipemic sample	7	0.7	Collection under non-fasting state? hyperlipidemia
Incorrect identification	44	4.6	Lack of knowledge/lax attitude of the phlebotomists/staff involved in sample collection and test ordering
Empty tube	10	1.0	
Requisition slip without sample	16	1.7	
Illegible hand writing	69	7.2	
Tube broken in the centrifuge	6	0.6	Improper centrifugation technique
Physician's request order missed	4	0.4	Carelessness at the level of the laboratory staff
Total	736	77.1	
Analytical			
Non-conformity with QC	6	0.6	Old QC, improper storage
Random error	15	1.6	Unknown cause
Calibration drift	10	1.0	Reagent instability, reagent changeover, expiry of calibration data with time
Reagent contamination	8	0.4	Reagent mix up, improper storage
Systemic error: probe, lamp, blocked tubing	36	3.8	Inherent technical problem/ routine wear and tear
Total	75	7.9	
Postanalytical			
Transcription errors	112	11.7	Manual error in copying numerical data from the printout on to the requisition slips
Prolonged turn around time	31	3.2	Problem arising due to inadequate water supply, irregular electricity, instrument problem

Clin Chem Lab Med 2010;48(1):63–66.

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Pre-analytical phase errors constitut majority of errors in clinical laborat

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Abstract

Objectives: Clinical laboratory errors pose a threat to patient safety and previous studies have demonstrated that pre-analytical error is the most common error type. Our study aimed to determine the types and frequency of errors occurring in clinical laboratory testing in contemporary practice.

Methods: Errors occurring in a core laboratory between 01/2022 and 05/2023 were recorded retrospectively. Errors were quantified using multiple data-streams including real-time manual technologist intervention, incidence reports filed by hospital staff/physicians, and retrospective assessment using automated reports from the lab information system (LIS). Errors were adjudicated and binned into pre-analytical, analytical, and post-analytical phases. Total test volumes were assessed in the LIS and electronic medical record.

Results: There were 37,680,242 billable results reported from approximately 11,000,000 specimens during the study period. In total, 87,317 errors occurred impacting 0.23 % (2,300 ppm) of billable results and approximately 0.79 % (7,900 ppm) of specimens. Among these errors, 85,894 (98.4 %, 984,000 ppm) were in the pre-analytical, 451 (0.5 %, 5,000 ppm) were in the analytical, and 972 (1.1 %, 11,000 ppm) occurred in the post-analytical phase. Hemolysis impacting specimen integrity (60,748/87,317, 69.6 %, 696,000 ppm) was the most common error. When excluding hemolysis, there were 26,569 errors documented (0.06 %, 600 ppm of billable results), among which 94.6 %, 1.7 % (17,000 ppm) and 3.7 % (37,000 ppm) were in the analytical phase respect

Conclusions: Observed previous studies with pr errors. High prevalence need for enhanced tools the pre-analytical phase

Keywords: laboratory er hemolysis; specimen int

Introduction

Clinical laboratory tes health care providers regarding disease scre prognosis. Thus, errors testing process (TTP) ma care [1, 2] and identifica practice improvement t rors may occur during post-analytical phases of can be divided into the the test is ordered [4] a referring to the process loading the specimen o cludes patient preparat transportation, storage phase is defined as the pr performed. Post-analytic after the testing process of improper result inter

Despite efforts by lal the analytical phase of t found that most errors accounting for approxi rors [8–14]. However, relatively small with mo oratory events [10]. The regarding documentation and collection of error, with several requiring physician documentation [11, 15], some assessing incidence reports [16] and others retrospectively

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Table 1: Distribution of laboratory errors in total testing process quality indicators.

Phases	Category	Counts	F	
Pre-analytical	Specimen integrity	85,133		
	Misidentification errors	761		
	Total	85,894		
Analytical	Interfering substance	279		
	QC out of range	65		
	Instrument problem	58		
	Reagent issues	49		
	Total	451		
Post-analytical	Comment errors	581		
	Test transcription errors	216		
	Comment errors with potential impact on patient care	131		
	Dilution errors	44		
	Total	972	1.1	11,000
Total		87,317	100	

Table 2: Breakdown of pre-analytical specimen integrity errors.

Category	Counts	Frequency, %	Frequency, ppm
Hemolyzed reported	41,047	48.2	482,000
Hemolyzed masked	19,701	23.1	231,000
Incorrect fill level	8,068	9.5	95,000
Clotted samples	5,840	6.9	69,000
Unsuitable samples for storage problems	5,780	6.8	68,000
Unsuitable samples for transportation problems	1,502	1.8	18,000
Not on ice	1,369	1.6	16,000
IV contamination	1,122	1.3	13,000
Too old to test	550	0.6	6,000
Sample integrity	92	0.1	1,000
Requisition errors	62	0.1	1,000
Total	85,133	100	N/A

During the study period, 4,993 proficiency tests were performed with a pass rate of 99.2 % (992,000 ppm, 4,955 out of 4,993). Turnaround time targets were met 172 out of 384 (44.8 % or 448,000 ppm) times throughout the study period (Supplementary Figure 1). The set goal of >99 % of critical results called to providers in ≤30 min was met for 16 out of 16 months (Supplementary Figure 2).

Supplementary Figure 1 List of turnaround time (TAT) and set goal for 50%, 90% and 97.5% broken down by months during the study period. Green and yellow shaded indicate that the TAT meet and did not meet the set goal respectively.

ALL	Goal	Jan-22	Feb-22	Mar-22	Apr-22	May-22	Jun-22	Jul-22	Aug-22	Sep-22	Oct-22	Nov-22	Dec-22	Jan-23	Feb-23	Mar-23	Apr-23
CBC																	
50%	None	0:10	0:10	0:10	0:10	0:10	0:10	0:10	0:10	0:10	0:10	0:11	0:11	0:10	0:11	0:10	0:10
90%	0:30	0:31	0:32	0:33	0:35	0:32	0:31	0:34	0:33	0:33	0:36	0:38	0:35	0:33	0:35	0:33	0:36
97.5%	None	0:56	0:56	0:59	1:06	1:00	0:54	0:57	0:53	0:57	1:06	1:06	0:58	0:54	1:00	0:58	1:01
CMP																	
50%	0:40	0:29	0:29	0:29	0:29	0:31	0:31	0:31	0:31	0:31	0:31	0:31	0:31	0:31	0:31	0:31	0:31
90%	0:60	0:43	0:46	0:45	0:45	0:48	0:48	0:49	0:49	0:49	0:49	0:51	0:50	0:48	0:52	0:51	0:50
97.5%	0:60	1:09	1:10	1:10	1:10	1:16	1:17	1:17	1:19	1:17	1:17	1:25	1:17	1:15	1:24	1:23	1:20
BMP																	
50%	0:40	0:28	0:28	0:28	0:28	0:30	0:30	0:30	0:30	0:30	0:30	0:30	0:30	0:30	0:30	0:30	0:30
90%	0:60	0:40	0:43	0:42	0:43	0:44	0:46	0:45	0:45	0:47	0:46	0:46	0:47	0:45	0:48	0:48	0:47
97.5%	0:60	1:05	1:05	1:06	1:06	1:09	1:12	1:09	1:12	1:13	1:13	1:16	1:12	1:11	1:22	1:19	1:16
Blood Gas																	
50%	0:05	0:06	0:06	0:06	0:05	0:06	0:06	0:06	0:06	0:06	0:06	0:06	0:05	0:05	0:05	0:05	0:05
90%	0:10	0:12	0:13	0:12	0:12	0:12	0:12	0:12	0:13	0:13	0:13	0:12	0:12	0:12	0:11	0:11	0:11
97.5%	0:10	0:19	0:19	0:17	0:20	0:19	0:20	0:17	0:20	0:20	0:21	0:17	0:19	0:18	0:17	0:17	0:17
Troponin																	
50%	0:40	0:37	0:36	0:36	0:36	0:37	0:38	0:37	0:38	0:38	0:38	0:37	0:37	0:37	0:37	0:38	0:37
90%	0:60	0:57	0:52	0:51	0:49	0:50	0:55	0:54	0:53	0:53	0:56	0:53	0:52	0:49	0:52	0:51	0:49
97.5%	0:60	1:14	1:10	1:08	1:04	1:04	1:13	1:09	1:10	1:07	1:11	1:10	1:06	1:03	1:11	1:06	1:03
NT-proBNP																	
50%	0:40	0:36	0:30	0:30	0:31	0:32	0:33	0:32	0:32	0:32	0:32	0:35	0:35	0:34	0:36	0:34	0:35
90%	0:60	1:13	1:02	0:58	1:00	1:04	1:07	0:59	1:07	1:04	1:06	1:12	1:09	1:07	1:42	1:19	1:19
97.5%	0:60	1:59	1:40	1:33	1:36	1:38	1:48	1:41	1:42	1:34	1:46	1:52	1:57	1:48	3:21	2:15	2:08
hCG																	
50%	0:20	0:19	0:22	0:18	0:19	0:15	0:16	0:22	0:16	0:19	0:22	0:18	0:17	0:18	0:17	0:18	0:20
90%	0:30	0:41	0:49	0:45	0:45	0:31	0:36	0:41	0:37	0:51	0:45	0:38	0:38	0:38	0:34	0:41	0:38
97.5%	0:30	0:53	1:18	0:54	1:03	0:52	0:58	1:04	1:03	1:10	0:56	0:50	0:58	0:47	0:38	1:12	0:52
UDS																	
50%	0:40	0:30	0:30	0:31	0:31	0:32	0:32	0:32	0:32	0:32	0:33	0:33	0:33	0:33	0:33	0:33	0:33
90%	0:60	0:53	0:51	0:54	0:50	0:52	0:52	0:54	0:53	0:53	0:56	1:06	0:57	0:56	0:57	0:55	0:59
97.5%	0:60	1:27	1:21	1:24	1:21	1:23	1:17	1:21	1:21	1:14	1:21	1:53	1:28	1:25	1:26	1:33	1:32
UA Macro																	
50%	None	0:12	0:12	0:12	0:11	0:10	0:11	0:10	0:12	0:11	0:13	0:12	0:12	0:33	0:33	0:33	0:33
90%	0:30	0:31	0:32	0:31	0:29	0:29	0:27	0:26	0:28	0:27	0:32	0:29	0:30	0:56	0:57	0:55	0:59
97.5%	None	0:51	0:56	0:52	0:50	0:48	0:46	0:44	0:46	0:45	0:56	0:47	0:52	1:25	1:26	1:33	1:32
PTT																	
50%	None	0:23	0:24	0:22	0:23	0:22	0:24	0:23	0:24	0:15	0:25	0:25	0:23	0:23	0:23	0:23	0:23
90%	0:45	0:36	0:35	0:33	0:34	0:34	0:34	0:34	0:36	0:30	0:36	0:37	0:36	0:36	0:36	0:34	0:36
97.5%	None	0:56	0:52	0:54	0:52	0:55	0:54	0:53	0:55	0:48	0:59	0:58	0:57	0:58	0:58	1:00	1:09

① 검사전단계 오류란

② 관련 사례들

③ 대응 지침들

정리

① 검사전단계 오류란

② 관련 사례들

③ 대응 지침들

정리

proposed pre-analytical quality specifications from IFCC

test order

input of request

ID collection handling

	Performance level			
	Optimum	Desirable	Minimum	Unacceptable
Pre-analytical Quality Specifications				
% requests with clinical question from general practitioners/total number of requests from general practitioners	>87	58-87	29-57	<29
% appropriate requests, with respect of clinical question from general practitioners /number of requests that reports clinical question from general practitioners	>97	65-97	32-64	<32
% requests without physician identification/total number of requests	<5	5.0-6.0	6.1-8.0	>8.0
% unintelligible requests/total number of requests	<0.2	0.20-0.25	0.26-0.30	>0.30
% requests with errors concerning patient identification/total number of requests	<0.4	0.40-0.50	0.51-0.60	>0.60
% requests with errors concerning physician identification/total number of requests		<0.1		
% requests with errors concerning input of tests (missing)/total number of requests	<0.3	0.30-0.40	0.41-0.50	>0.50
% requests with errors concerning input of tests (added)/total number of requests		<0.1		
% requests with errors concerning input of tests (misinterpreted)/total number of requests	<0.2	0.20-0.25	0.26-0.30	>0.30
% samples lost-not received/total number of samples	<0.2	0.20-0.40	0.41-0.60	>0.60
% samples collected in inappropriate container/total number of samples	<0.07	0.07-1.13	1.14-0.20	>0.20
% samples hemolysed (chemistry)/total number of samples	<1	1.0-1.5	1.6-2.0	>2.0
% samples clotted (hematology)/total number of samples with anticoagulant	<0.5	0.50-1.0	1.1-2.0	>2.1
% samples with insufficient sample volume/total number of samples	<0.4	0.40-0.80	0.81-1.20	>1.20
% samples with inadequate sample-anticoagulant/total number of samples with anticoagulant	<0.2	0.20-0.30	0.31-0.40	>0.40
% samples damaged in transport/total number of samples		<0.1		
% samples improperly labelled/total number of samples	<0.07	0.07-0.15	0.16-0.20	>0.20

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Guidelines and Recommendations

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Recommendations from the IFCC Working Group on Laboratory Errors and Patient Safety for the Global Adoption of an Essential Quality Indicators Panel in Laboratory Medicine

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Abstract

Objectives: To maximize participation in the international standardization effort, this recommendation aims to update the international guidance of the Working Group on Laboratory Errors and Patient Safety of the International Federation of Clinical Chemistry and Laboratory Medicine by identifying a limited and globally applicable panel of Essential Quality Indicators (QIs) focused on patient safety, clinical outcomes, and harmonization across medical laboratories.

Methods: Through a consensus meeting, experts from multiple countries reviewed the IFCC Model of Quality Indicators (MQI) to identify high-priority indicators. Selection prioritized the probability of patient harm, ease of detection, and feasibility of data collection and implementation within national contexts.

Results: Six essential QIs covering the total testing process were identified and ratified: (1) rate of misidentified requests (Pre-MisR) and misidentified samples (Pre-MisS); (2) rate of sample rejections (Pre-RejS); (3) rate of hemolysis detected either by automated hemolysis index (Pre-HemI) or visual inspection (Pre-HemV); (4) rate of unacceptable results in External Quality Assessment/Proficiency Testing (Intra-Unac); (5) turnaround time of cardiac troponin at the 90th percentile for the emergency room (Post-TnTAT, Post-TnTAT clin); and (6) rate of incorrect laboratory reports (Post-RectRep). Recommendations on calculation, reporting frequency, and integration into IFCC and national comparison programs are provided.

Conclusions: The proposed essential QI panel provides a standardized and feasible framework to support its

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Table 1: List of the essential Quality Indicators.

	QI	Phase	Calculation	IFCC
#1	Rate of misidentified requests	Pre	%: Number of misidentified requests/Total number of requests	Pre-M
	Rate of misidentified samples		%: Number of misidentified samples/Total number of samples	Pre-M
#2	Rate of samples rejection	Pre	%: Number of samples rejected/Total number of samples	Pre-R
#3	Rate of hemolysis	Pre	%: Number of samples with free hemoglobin>0.50 g/L/Number of checked samples for hemolysis	Pre-H mate Pre-H inspe
#4	Rate of unacceptable performance in EQA-PT schemes per year	Analytical	%: Number of unacceptable performances in EQA-PT schemes in a year/ Total number performances in EQA schemes in a year	Intra-
#5	90th percentile of turnaround time of troponin (analytical)	Pre analytical	90th percentile of TAT (min) of cardiac troponin for the ER: From sample reception in the lab to release of results	Post-
	90th percentile of turnaround time of troponin (clinical)	Post	90th percentile of TAT (min) of cardiac troponin for the ER: From blood sampling to release of results	Post- (new)
#6	Rate of corrected laboratory reports	Post	%: Number of corrected reports by laboratory after release/Total number of reports (requests)	Post-

Pre-MisR, misidentified requests; Pre-MisS, misidentified samples; Pre-RejS, rate of sample rejections; Pre-HemI, rate of hemolysis detecte hemolysis index; Pre-HemV, rate of hemolysis detected by visual inspection; Intra-Unac, rate of unacceptable results in External Quality A Proficiency Testing; Post-TnTAT, Post-TnTAT clin, turnaround time of cardiac troponin at the 90th percentile for the emergency room; Post- incorrect laboratory reports.

Table 2: Recommended frequency of evaluation to support standardized implementation of the essential QIs in the comparison programs.

	QI	IFCC QIs	Evaluation schedules
#1	Rate of misidentified requests	Pre-MisR	Data collection: Everyday for a month
	Rate of misidentified samples	Pre-MisS	April – August – December
#2	Rate of samples rejection	Pre-RejS (new)	
#3	Rate of hemolysis	Pre-HemI (automated index)	
		Pre-HemV (Visual inspection)	
#5	90th percentile of turnaround time of troponin (analytical)	Post-TnTAT	
	90th percentile of turnaround time of troponin (clinical)	Post-TnTATclin (new)	
#6	Rate of corrected laboratory reports	Post-RectRep	
#4	Rate of unacceptable performance in EQA-PT schemes per year	Intra-Unac	Data collection: Every year

Pre-MisR, misidentified requests; Pre-MisS, misidentified samples; Pre-RejS, rate of sample rejections; Pre-HemI, rate of hemolysis detected by automated hemolysis index; Pre-HemV, rate of hemolysis detected by visual inspection; Intra-Unac, rate of unacceptable results in External Quality Assessment/Proficiency Testing; Post-TnTAT, Post-TnTAT clin, turnaround time of cardiac troponin at the 90th percentile for the emergency room; Post-RectRep, rate of incorrect laboratory reports.

No	검사시행여부	검사명	결과	2025년				
1	시행	**총 검체 수**	입력값	518,284	일반화학, 일반혈액, 혈액응고			
2	시행	**검사 취소된 총 검체 수**	입력값	202	측정부적합현황	처리사항	검사취소	
3	미시행	검사취소율(접수취소율, 검체)	계산값					
4	미시행	-총 검체 수	상위검사 입력값 사용					
5	미시행	-검사 취소된 총 검체 수						
6	시행	환자확인 오류율(검체)						
7	시행	**-환자확인 오류 검체 수**	입력값	25	1 채혈(바코드 불일치)	취소사유	측정부적합현황	
8	시행	-총 검체 수			24 채혈(환자 바뀜)	취소사유	측정부적합현황	
9	시행	-검사 취소된 총 검체 수						
10	시행	검체 종류 오류율(검체)						
11	시행	**-부적절한 검체 종류(기질, matrix) 검체 수**	입력값	26	0 검체(Abscess)	취소사유	측정부적합현황	
12	시행	-검사 취소된 총 검체 수			0 검체(Pus)	취소사유	측정부적합현황	
13	시행	-총 검체 수			10 검체(검체지정 오류)	취소사유	측정부적합현황	
14	시행	응기오류율			11 검체(오염)	취소사유	측정부적합현황	
					5 검체(희석)	취소사유	측정부적합현황	
15	시행	**-응기오류 검체 수**	입력값	40	검체(부적합 응기)	취소사유	측정부적합현황	
16	시행	-검사 취소된 총 검체 수						
17	시행	-총 검체 수						
18	시행	용량부족 검체비율						
19	시행	**-용량부족 검체 수**	입력값	353	검체(양부족)	취소사유	측정부적합현황	
20	시행	-검사 취소된 총 검체 수						
21	시행	-총 검체 수						
22	시행	운반 중 파손된 검체 비율						
23	시행	**-운반 중 파손된 검체 수**	입력값	5	검체(응기 파손)	취소사유	측정부적합현황	
24	시행	-총 검체 수						
25	시행	-검사 취소된 총 검체 수						
26	미시행	운반 지연 검체 비율						
27	미시행	**-검체 채취 후 검사실접수까지의 시간이 지연된 검체 수(각 검사별 정해진 기준보다 지연된 경우)**	입력값					
28	미시행	-총 검체 수						
29	미시행	-검사 취소된 총 검체 수						
30	미시행	용혈(free Hb>0.5 g/L) 일반화학검사 비율(육안,검체)						
31	미시행	**-육안으로 확인된 일반화학검사 용혈 검체 수(용혈 기준은 free Hb>0.5 g/L로 정함)**	입력값					
32	미시행	**-검사 취소된 일반화학검사 검체 수**	입력값					
33	미시행	**-일반화학검사 검체 수**	입력값					
34	시행	용혈(free Hb>0.5 g/L) 일반화학검사 비율(장비,검체)						
35	시행	**-장비의 H-index로 확인된 일반화학검사 용혈 검체 수(용혈 기준은 free Hb>0.5 g/L로 정함)**	입력값	906	L30+L80 H index > 50	Flag	검체비고 입력내역 조회	
36	시행	**-검사 취소된 일반화학검사 검체 수**	입력값	130	검사취소	처리사항	측정부적합현황	
37	시행	**-일반화학검사 검체 수**	입력값	314,461	L30+L80	검체	검사 검체 통계	
38	미시행	용혈(free Hb>0.5 g/L) 일반화학검사 검체거부율(검체)						
39	미시행	**-용혈로 검체를 거부한(재채혈을 요구한) 일반화학 검체 수**	입력값	701	검체(Hemolysis)	취소사유	측정부적합현황	
40	미시행	**-검사 취소된 일반화학검사 검체 수**	입력값	0	검체(Hemolysis)-검사취소	취소사유-처리사항	측정부적합현황	
41	미시행	**-일반화학검사 검체 수**	입력값	314,461	L30+L80	검체	검사 검체 통계	
42	시행	응괴가 발견된 일반혈액검사(CBC) 비율(검체)						
43	시행	**-응괴가 발견된 일반혈액검사(CBC)검체 수**	입력값	506	검체(Clot)	취소사유	측정부적합현황	
44	시행	**-검사 취소된 일반혈액검사(CBC) 검체 수**	입력값	46	검사취소	처리사항	측정부적합현황	
45	시행	**-일반혈액검사(CBC) 검체 수**	입력값	160,796	L20	검체	검사 검체 통계	
46	시행	항응고제 비율이 적절하지 않은 응고검사 비율(검체)						
47	시행	**-항응고제 비율이 적절하지 않은 응고검사 검체 수**	입력값	40	검체(양과다), 검체(양부족)	취소사유	측정부적합현황	
48	시행	**-검사 취소된 응고검사 검체 수**	입력값	26	검사취소	처리사항	측정부적합현황	
49	시행	**-응고검사 검체 수**	입력값	43,027	L22	검체	검사 검체 통계	

2026 년 01 차 || 검사 전 단계 품질지표 (시범)

최종전송일자 : 2026 년 09 월 16 일
작성자 : 류윤호

시행여부	검사명[단위]	검사 전 단계 품질지표 분항 (검체)
시행	총 검체 수 [ ]	518284
시행	검사 취소된 총 검체 수 [ ]	202
미시행	검사취소율(접수취소율, 검체) [백분율(%)]	
미시행	-총 검체 수 [ ]	
미시행	-검사 취소된 총 검체 수 [ ]	
시행	환자확인 오류율(검체) [백분율(%)]	
시행	-환자확인 오류 검체 수 [ ]	25
시행	-총 검체 수 [ ]	518284
시행	-검사 취소된 총 검체 수 [ ]	202
시행	검체 종류 오류율(검체) [백분율(%)]	
시행	-부적절한 검체 종류(기질, matrix) 검체 수 [ ]	26
시행	-검사 취소된 총 검체 수 [ ]	202
시행	-총 검체 수 [ ]	518284
시행	응기오류율 [백분율(%)]	
시행	-응기오류 검체 수 [ ]	40
시행	-검사 취소된 총 검체 수 [ ]	202
시행	-총 검체 수 [ ]	518284
시행	용량부족 검체비율 [백분율(%)]	
시행	-용량부족 검체 수 [ ]	353
시행	-검사 취소된 총 검체 수 [ ]	202
시행	-총 검체 수 [ ]	518284
시행	운반 중 파손된 검체 비율 [백분율(%)]	
시행	-운반 중 파손된 검체 수 [ ]	5
시행	-총 검체 수 [ ]	518284
시행	-검사 취소된 총 검체 수 [ ]	202
미시행	운반 지연 검체 비율 [백분율(%)]	
미시행	-검체 채취 후 검사실접수까지의 시간이 지연된 검체 수(각 검사별 정해진 기준보다 지연된 경우) [ ]	
미시행	-총 검체 수 [ ]	
미시행	-검사 취소된 총 검체 수 [ ]	
미시행	용혈(free Hb)>0.5 g/L) 일반화학검사 비율(육안,검체) [백분율(%)]	
미시행	-육안으로 확인된 일반화학검사 용혈 검체 수(용혈 기준은 free Hb)>0.5 g/L로 정함) [ ]	
미시행	-검사 취소된 일반화학검사 검체 수 [ ]	
미시행	-일반화학검사 검체 수 [ ]	
시행	용혈(free Hb)>0.5 g/L) 일반화학검사 비율(장비,검체) [백분율(%)]	
시행	-장비의 H-index로 확인된 일반화학검사 용혈 검체 수(용혈 기준은 free Hb)>0.5 g/L로 정함) [ ]	906
시행	-검사 취소된 일반화학검사 검체 수 [ ]	130
시행	-일반화학검사 검체 수 [ ]	314461

시행	용혈(free Hb)>0.5 g/L) 일반화학검사 검체거부율(검체) [백분율(%)]	
시행	-용혈로 검체를 거부한(재채혈을 요구한) 일반화학 검체 수 [ ]	701
시행	-검사 취소된 일반화학검사 검체 수 [ ]	130
시행	-일반화학검사 검체 수 [ ]	314461
시행	응괴가 발견된 일반혈액검사(CBC) 비율(검체) [백분율(%)]	
시행	-응괴가 발견된 일반혈액검사(CBC)검체 수 [ ]	506
시행	-검사 취소된 일반혈액검사(CBC) 검체 수 [ ]	46
시행	-일반혈액검사(CBC) 검체 수 [ ]	160796
시행	항응고제 비율이 적절하지 않은 응고검사 비율(검체) [백분율(%)]	
시행	-항응고제 비율이 적절하지 않은 응고검사 검체 수 [ ]	40
시행	-검사 취소된 응고검사 검체 수 [ ]	26
시행	-응고검사 검체 수 [ ]	43027

EFLM Paper

Pieter Vermeersch*, Glynis Frans, Alexander von Meyer, Seán Costelloe, Giuseppe Lippi and Ana-Maria Simundic

How to meet ISO15189:2012 pre-analytical requirements in clinical laboratories? A consensus document by the EFLM WG-PRE

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Abstract: The International Organization for Standardization (ISO) 15189:2012 standard aims to improve quality in medical laboratories through standardization of all key elements in the total testing process, including the pre-analytical phase. It is hence essential that accreditation bodies, assessing laboratories against ISO15189:2012, pay sufficient attention to auditing pre-analytical activities. However, there are significant differences in how technical auditors interpret the pre-analytical requirements described in ISO15189:2012. In this consensus document, the European Federation of Clinical Chemistry and Laboratory Medicine (EFLM) Working Group for Pre-analytical Phase (WG-PRE) sets out to review pre-analytical requirements contained in ISO15189:2012 and provide guidance for laboratories on how to meet these requirements. The target audience for this consensus document is laboratory professionals who wish to improve the quality of the pre-analytical phase in their laboratory. For each of the ISO requirements described in ISO15189:2012, members of EFLM WG-PRE agreed by consensus on minimal recommendations and best-in-class solutions. The minimal consensus recommendation was defined as the minimal specification which laboratories should implement in their quality management system to adequately address the pre-analytical requirement described in ISO15189:2012. The best-in-class solution describes the current state-of-the-art in fulfilling a particular pre-analytical requirement in ISO15189:2012. We fully acknowledge that not every laboratory has the means to implement these best-in-class solutions, but we hope to challenge laboratories in critically evaluating and improving their current procedures by providing this expanded guidance.

Keywords: accreditation; ISO15189; pre-analytical phase; quality improvement.

Pieter Vermeersch and Glynis Frans contributed equally to this work.

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Introduction

The total testing process (TTP) is a complex concept in laboratory medicine, originally referred to as the "brain-to-brain" cycle by George Lundberg nearly 50 years ago [1]. The TTP can be summarized as follows: the physician orders the necessary test prescriptions; the patient is identified and prepared for sampling; the samples are collected, transported, identified, stored, and prepared for analysis. All the processes prior to actual analysis, both outside and inside the laboratory, comprise the pre-analytical phase. After analysis, test results are validated, reported, and interpreted by laboratory specialists and requesting physicians, who take further medical decisions based on test results [1].

The overall frequency of errors in laboratory medicine is approximately 0.3%, which remains lower than those of other medical diagnostic disciplines (e.g., histopathology, with an error rate of nearly 5.0%) [2]. The pre-analytical

Table 1: ISO15189:2012 requirements and corresponding EFLM WG-PRE recommendations/solutions relating to quality management of the pre-examination phase.

ISO paragraph	Question	ISO requirement	Minimal recommendation	Grade	Best-in-class solution	Grade
4.1.2.4 – Quality objectives and planning	How to define quality objectives and quality indicators for pre-examination processes?	The quality objectives shall be measurable and consistent with the quality policy.	Laboratories should at least monitor one of the following quality indicators: number of misidentification errors, test transcription errors, incorrect sample types, insufficiently filled samples, unsuitable samples, contaminated samples, hemolyzed samples, or clotted samples.	2a	Pre-analytical quality indicators are monitored according to framework provided by the IFCC Model of Quality Pre-analytical Indicators. Laboratories should implement all quality indicators that are relevant for their setting based on risk-assessment. Participation in the IFCC External Quality Assessment program is encouraged.	2a
	How frequent should pre-analytical quality objectives/quality indicators be evaluated?	Planning of the quality management systems is carried out to meet the requirements and the quality objectives.	Yearly.	1	Frequency according to the framework provided by the IFCC Model of Quality Pre-analytical Indicators.	2a
4.1.2.5 – Responsibility, authority and interrelationships	Is it required to appoint a person in the laboratory who is responsible for the pre-examination phase? Definition, requirements?	Yes. Appointment of person(s) responsible for each laboratory function and appointment of deputies for key managerial and technical personnel.	A dedicated laboratory staff member should be appointed who is responsible for all pre-examination aspects both within and outside of the central laboratory.	1	A dedicated laboratory medicine specialist should be appointed who is responsible for all pre-examination aspects both within and outside of the central laboratory.	3
4.1.2.6 – Communication processes	Should notifications and problems concerning the pre-analytical phase be communicated to hospital and/or laboratory personnel? How should this be done?	Yes.	Any deterioration of the quality objectives or significant remarks obtained during internal audits shall be communicated and documented.	1	At least half-yearly meetings with all stakeholders to discuss expectations, internal audit results (if performed during that period), quality indicators and non-conformities. This should be done using a well-developed communication plan.	3
4.8 – Resolution of complaints **4.9 – Identification and control of nonconformities**	How should complaints and non-conformities about the pre-examination processes be handled?	Should be handled similar to other complaints, non-conformities, etc. including documentation.	Complaints of the pre-analytical phase should be handled in accordance with the general laboratory policy.	1	–	
4.10 and 4.11 – Corrective and preventive actions	How should corrective and preventive actions concerning pre-examination processes be handled?					

definition of laboratory errors

- ISO/TS 22367
  - failure of planned action to be completed as intended,
  - or use a wrong plan to achieve an aim,



~~ISO/TS 22367:2008~~

Medical laboratories — Reduction of error through risk management and continual improvement

Withdrawn (Edition 1, 2008)

→ New version available: ISO 22367:2020

Life cycle

Previously

Withdrawn
ISO 22367:2020

Now

Published
ISO 22367:2026
Stage: **60.60**

ISO 22367:2026 - 3 Terms and
Gemini에게 물어보기
iso.org/standard/87738.html

Preview ISO 22367:2026

ISO 22367:2026

Medical laboratories — Application of risk manag...

Published (Edition 2, 2026)

Table of Contents

- Foreword
- Introduction
- 1 Scope
- 2 Normative references
- 3 Terms and definitions
- 4 Risk management
- 5 Proactive risk management
- 6 Proactive risk analysis
- 7 Risk evaluation
- 8 Risk control
- 9 Risk management review
- 10 Risk monitoring, analysis and control ...
- 11 Immediate actions to reduce risk
- Annex A Implementation of risk manage...
- Annex B Guidance on establishing risk a...

Figures

Tables

Rendering disclaimer

Languages: **en** fr

Search in document

3.42
use error

laboratory medicine user (3.43) action or lack of user action while performing a laboratory examination (3.3) or using an IVD medical device (3.11) or performing any task in any procedure (3.19) that leads to a different result than that intended by the laboratory or manufacturer or expected by the user

Note 1 to entry: Use error includes the inability of the user to complete a task.

Note 2 to entry: Use errors can result from a mismatch between the characteristics of the user, user interface, task, or use environment.

Note 3 to entry: Users might be aware or unaware that the use error has occurred.

Note 4 to entry: An unexpected physiological response of the patient is not by itself considered use error.

Note 5 to entry: A malfunction of an IVD medical device that causes an unexpected result is not considered a use error.

Note 6 to entry: Use error includes the use of an examination result for an unintended target group or for an unintended diagnostic or patient management purpose.

Note 7 to entry: The term was chosen over "user error", "human error" or "laboratory error" because not all causes of error are partially or solely due to the user. Use errors are often the result of poorly designed user interface or processes (3.21), or, inadequate instructions for use (3.14).

[SOURCE: IEC 62366-1:2015, 3.21 modified — In the definition, "user" was changed to "laboratory medicine user", and "performing a laboratory examination" and "performing any task in any procedure" were added; the original Note 6 to entry was removed and a new Note 6 to entry and a Note 7 were added.]

services of a medical laboratory, including the pre-examination, examination, and post-

Technical Committee : ISO/TC 212

06:53

오류란 무엇인가?

Definition of laboratory error — ISO/TS 22367:2008, 3.1

Traditional definition

Laboratory error

“planned action fails”

“wrong plan is used”

검사 의뢰부터 결과 보고,
해석과 후속 대응까지
검사실 주기의 어느 단계에서든 발생

References: ISO/TS 22367:2008, 3.1; Plebani M. Ann Clin Biochem. 2010;47:101–110.

정의의 업데이트

From "laboratory error" to "use error" — ISO 22367:2020/2026

2008

Laboratory error

- planned action fails
- wrong plan is used
- any part of the laboratory cycle

오류를 "행위/계획의 실패"로 정의

2020 / 2026

Use error

사용자의 행위 또는 행위 누락으로 인해
검사실·제조자·사용자가
의도하거나 **기대**한 것과 **다른 결과**가 발생하는 것

핵심은 개인 비난보다 시스템·절차·환경의 불일치 탐지

오류는 "누가 잘못했는가?"보다 "어떤 과정이 이 결과를 허용했는가?"를 묻는 출발점이다.

References: ISO 22367:2026, 3.42; ISO 22367:2020, 3.41; ISO 22367:2026 abstract.

① 검사전단계 오류란

② 관련 사례들

③ 대응 지침들

정리

검체채혈
2026-08-31
2026-08-30
2026-08-30

응급화학 접수지연 분석 보고서

2025.10 ~ 2026.08 · 진단검사의학과

1. 배경 및 방법

임상의 불만: 입원환자 응급화학이 일반혈액보다 늦게 접수

기준선 30 분 (SST 원심 10 분 + TLA 부재 수기처리)

대상: 입원환자 중 환자번호 + 채혈시각(≤3 초) 매칭 9,681 쌍 (응급실 제외)

기간: 2025-10-06 ~ 2026-08-31 (약 11 개월, 매주 월요일)

지표: 응급화학 접수 − 일반혈액 접수(분)

2. 전체 결과

응급화학 지연비율 80.7% (7,809 / 9,681)

중앙값 +15.0 분 / 평균 +19.7 분

3. 월별 결과 (30 분 기준)

월	n	최소	중간	최대	평균	기준
25-10	695	-59.4	26.0	279.3	30.8	초과(평균)
25-11	886	-26.9	12.4	162.6	12.2	-
25-12	1091	-41.0	8.5	52.4	7.7	-
26-01	854	-52.2	6.3	160.6	8.6	-
26-02	759	-26.4	12.0	51.7	9.5	-
26-03	935	-53.7	11.9	50.5	10.8	-
26-04	858	-286.6	15.8	193.0	25.8	-
26-05	740	-16.6	13.4	45.3	12.8	-
26-06	1031	-84.4	16.8	168.6	19.1	-
26-07	876	-59.7	20.8	129.4	28.2	-
26-08	956	-73.1	**39.7**	180.2	**52.0**	초과

4. 시간대별 결과 (0~23 시)

시	n	최소	중간	최대	평균	기준
0	15	8	12.7	18.2	12.7	-
1	16	3	17.2	24	16.7	-
2	15	0.2	14.8	30.8	14.9	-
3	23	-3.8	8	32.2	11.8	-
4	780	-11.8	19.1	57	19	-
5	1143	-58.5	12.8	69	14.5	-
6	1250	-53.7	11.6	77.4	13.9	-
7	1658	-84.4	12.5	149.2	13.5	-
8	1638	-286.6	13.5	162.6	17.6	-
9	1191	-59.4	**30**	150.8	**40.1**	초과
10	363	-12.6	29.8	279.3	57.8	초과(평균)
11	172	-15	13.5	180.2	24.1	-
12	109	-16.2	14.9	41.3	12.4	-
13	147	-30.6	11.7	33.6	8.7	-
14	210	-73.1	12.2	32	9.5	-
15	186	-16.2	13.8	51.7	11.4	-
16	264	-19.8	11.9	36.9	9.3	-
17	187	-16.8	13.8	45.3	12.6	-
18	78	-6.8	18.3	52.4	19.5	-
19	106	-6.4	16.6	91.1	16.5	-
20	58	-5.1	16.8	51.9	16.6	-
21	25	4.4	17	143.4	22.7	-
22	28	-8.8	14.1	42.9	15.1	-
23	19	2.7	13.3	26.9	14.3	-

5. 월별 피크시간대 (n≥20)

월	피크시간	n	중앙값
25-10	10 시	48	**55.3**
25-11	4 시	67	22.8
25-12	17 시	22	18.3
26-01	10 시	26	14.4
26-02	9 시	62	18.3
26-03	5 시	132	19.5
26-04	10 시	41	**107.6**
26-05	9 시	83	18.4
26-06	9 시	165	33.1
26-07	9 시	197	**50.9**
26-08	10 시	115	**107.8**

6. 결론 및 제언

① 8 월만 월 단위 기준초과 (중앙값 39.7 분 / 평균 52.0 분)

② 9~10 시대 집중 초과, 4 월·8 월 10 시 극단 지연(100 분↑)

③ 접수 처리 순서 · 원심분리 대기열 · 인력 배치 확인 권고

④ 우선 검토 순서: 8 월 전체 → 4 월 4 주

7. 데이터 정합성

접수<채혈 시각 역전 사례 0 건

검사일자 100% 월요일 일치

채혈시각 결측 시 라벨출력일시 대체, 매칭 영향 없음 확인

8-1. 주별 결과 (10~11 월)

주	n	최소	중간	최대	평균	기준
10/06	86	-0.8	18.0	51.5	17.9	-
10/13	200	-59.4	30.0	100.0	29.9	-
10/20	201	-4.8	17.2	279.3	26.7	-
10/27	208	-12.6	**41.7**	94.1	**40.8**	초과
11/03	232	-26.9	12.8	162.6	9.3	-
11/10	222	-15.8	11.2	60.6	14.8	-
11/17	209	-15.4	12.3	33.8	10.9	-
11/24	223	-10.0	13.2	143.4	13.8	-
12/01	210	-30.6	13.9	38.0	11.1	-
12/08	205	-13.6	7.2	29.4	7.4	-
12/15	230	-15.5	11.8	33.2	8.2	-
12/22	222	-9.5	7.5	52.4	5.9	-

8-2. 주별 결과 (12~2 월)

주	n	최소	중간	최대	평균	기준
12/29	224	-41.0	5.4	39.6	5.5	-
01/05	211	-13.2	5.9	160.6	12.3	-
01/12	218	-22.0	8.4	91.1	10.0	-
01/19	223	-52.2	2.7	36.3	4.3	-
01/26	202	-18.2	8.9	31.9	7.9	-
02/02	219	-15.4	15.8	40.0	14.5	-
02/09	216	-26.4	10.7	51.7	8.9	-
02/16	129	-25.4	15.1	36.9	7.8	-
02/23	195	-12.9	7.6	29.5	5.7	-
03/02	154	-46.0	9.0	50.5	10.3	-
03/09	184	-8.7	14.3	30.4	11.4	-
03/16	181	-53.7	15.0	44.8	12.1	-

8-3. 주별 결과 (3~5 월)

주	n	최소	중간	최대	평균	기준
03/23	226	-29.3	6.9	29.6	7.9	-
03/30	190	-11.1	14.0	40.8	12.8	-
04/06	204	-75.2	12.2	34.7	9.1	-
04/13	238	-3.5	19.3	56.9	20.3	-
04/20	198	-21.9	8.2	30.6	8.6	-
04/27	218	-286.6	**56.3**	193.0	**62.9**	초과
05/04	218	-10.7	17.0	45.3	14.9	-
05/11	205	-2.8	12.0	40.5	11.8	-
05/18	193	-7.0	17.3	29.9	16.0	-
05/25	124	-16.6	3.9	30.4	5.8	-
06/01	198	-14.9	17.6	34.0	17.7	-
06/08	200	-11.8	18.2	37.9	14.5	-

8-4. 주별 결과 (6~8 월)

주	n	최소	중간	최대	평균	기준
06/15	219	-58.5	15.1	96.2	22.0	-
06/22	217	-84.4	16.0	168.6	17.0	-
06/29	197	-13.6	15.3	72.4	24.3	-
07/06	245	-59.7	**38.5**	129.4	**42.4**	초과
07/13	207	-25.5	15.2	75.8	17.6	-
07/20	205	-16.8	27.1	88.1	32.7	초과(평균)
07/27	219	-39.8	13.7	55.9	18.2	-
08/03	230	-9.0	**47.8**	109.2	**55.2**	초과
08/10	210	-23.4	25.0	98.5	42.0	초과(평균)
08/17	126	-8.2	22.0	77.4	31.4	초과(평균)
08/24	201	-3.4	**56.5**	180.2	**70.8**	초과
08/31	189	-73.1	**34.8**	157.5	**52.8**	초과

작성일: 2026. 9. 1.

작성: 임상화학

종합

- 오류는 진단검사의학과의 전과정에서 발생할 수 있으며, 특히 검사전단계에서의 비도가 높다.
- 오류는 실패가 아닌, 의도한 바와 다른 결과론적 개념이므로, 개인의 책임이 아닌 검사실 전반의 체계를 투영하는 것이다.
- 오류의 발견과 근본원인 분석은 체계적인 검사실 운영의 주요 인자이다.

감사합니다.